

B MALULEKA ATTORNEYS INC.

Web Portal & AI-Driven Client Intake System

Standard: IREB (International Requirements Engineering Board)

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Certifications: IREB CPRE Foundation | ISTQB Foundation Level

Version: 1.0 | Date: July 2025 | Status: Final

Target Organisation: B Maluleka Attorneys Inc.

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1. Context & Elicitation Methodology

1.1 Elicitation Technique: Artifact-Based Elicitation

Per IREB guidelines, **Artifact-Based Elicitation** was utilised to define the functional baseline for the frontend system. This technique derives requirements by analysing existing comparable systems rather than relying solely on stakeholder interviews, reducing ambiguity and anchoring requirements to observable, verifiable artefacts.

Primary Reference Artifact	Competitor Web Portal — www.mnguniattorneysinc.co.za
Elicitation Process	Structure, navigation hierarchy, service presentation, and user flow of the reference artifact were

2. System Vision & Business Objectives

Primary Objective: Deliver a professional legal web portal for B Maluleka Attorneys modelled after industry standards, backed by an automated AI-driven client intake system.

Business Value Targets:

- Eliminate manual intake overhead through end-to-end automation of the client inquiry lifecycle.
- Achieve instant response times for routine legal FAQs via AI-generated replies.
- Ensure immediate Director escalation for high-value, urgent, or complex legal matters.
- Guarantee zero loss of client inquiry data through fault-tolerant database persistence.

3. Functional Requirements (FR)

3.1 Frontend Web Portal Requirements

FR-01	Structural Parity: The website SHALL replicate the page structure, navigation layout, and service breakdown of the benchmark artifact (www.mnguniattorneysinc.co.za).
FR-02	Custom Visual Branding:
FR-02.1	Colour Theme: The website SHALL utilise a custom Gold and Black visual theme across all pages, overriding the styling of the baseline artifact.
FR-02.2	Brand Identity: The official B Maluleka Attorneys logo SHALL be persistently displayed in the header and footer of every page.
FR-03	Client Intake Interface: The website SHALL feature a prominent, mobile-responsive client intake form containing the following mandatory input fields: Full Name (Text), Email Address (Email Format), Contact / Phone Number (Phone Format), and Inquiry / Message Body (Text Area). All fields are mandatory.
FR-04	Webhook Handshake: Upon form submission, the website SHALL transmit an asynchronous HTTP POST JSON payload to the designated Make.com webhook endpoint.

3.2 Automation Backend Requirements (Make.com Architecture)

FR-05	Payload Validation: The system SHALL evaluate incoming webhook payloads. If mandatory fields (Message or Contact Information) are missing or blank, the system SHALL terminate execution early to prevent corrupt data from reaching downstream modules.
FR-06	Centralised Database Persistence: The system SHALL log every validated intake payload to the central Google Sheets Database (B Maluleka Client Intake Log). Data insertion MUST occur prior to AI evaluation to guarantee fault tolerance in the event of downstream service timeouts or API failures.
FR-07	AI Intent Classification & RAG: The system SHALL pass the inquiry message to a Make AI Agent integrated with a dedicated Knowledge Base (RAG) containing firm guidelines and service details. The AI Agent SHALL evaluate the query and categorise it into one of two output states: FAQ_ROUTINE or REQUIRES_HUMAN_ATTENTION .
FR-08	Dynamic Multi-Branch Routing:
FR-08A	Path A — Is FAQ: If classified as FAQ_ROUTINE, the system SHALL trigger Gmail to send an automated, personalised response containing accurate domain information directly to the client's email address.
FR-08B	Path B — Needs Human Attention: If classified as REQUIRES_HUMAN_ATTENTION, the system SHALL trigger Gmail to send an immediate, high-priority alert containing full client details and inquiry summary to the Director's email address.

4. Non-Functional Requirements (NFR)

NFR ID	Category	Requirement
NFR-01	Fault Tolerance & Reliability	The database logging step must be decoupled from the AI engine so that third-party AI or email
NFR-02	Performance & Latency	The frontend intake form must receive an HTTP 200 OK webhook acknowledgment within < 2.0
NFR-03	Usability & Responsive Design	The frontend web interface SHALL fully adapt across mobile, tablet, and desktop viewports, con
NFR-04	Data Integrity	The automation backend SHALL sanitise input payloads to preserve multiline string formatting a

5. Requirements Traceability Matrix (RTM)

The RTM below maps each requirement to its elicitation source and the corresponding implementation component, ensuring full traceability from business need to technical delivery.

Req. ID	Description	Source / Elicitation Basis	Implementation Component
FR-01	Layout & Navigation	Artifact (Mnguni Attorneys Website)	Netlify Frontend Application
FR-02	Gold/Black Styling & Logo	Stakeholder Delta Requirement	Custom CSS Theme & Asset Headers
FR-03	Client Intake Form	Artifact + Stakeholder Objective	Frontend Form (HTML/JS)
FR-04 FR-05	Webhook & Payload Filtering	System Architecture Specification	Make.com Custom Webhook & Filter
FR-06	Intake Logging	Technical Integrity Requirement	Google Sheets API (Add a Row)
FR-07	AI Categorisation	Automation Solution Specification	Make AI Agent + Knowledge Base (RAG)
FR-08	FAQ vs Escalation Routing	Business Rule Specification	Make.com Router + Gmail API Modules
NFR-01	Fault Tolerance	Technical Integrity Requirement	Google Sheets logged before AI call
NFR-02	Performance & Latency	System Architecture Specification	Make.com Webhook Response + Pipeline SLA
NFR-03	Responsive Design	Stakeholder Objective	CSS Media Queries / Netlify Frontend
NFR-04	Data Integrity	Technical Integrity Requirement	Make.com Payload Sanitisation

Document Status	Final
Version	1.0

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